Supplementary information for OpenDC-LCA

Han Hu[a], Darin W. Nutter[a]

[a]*Department of Mechanical Engineering, University of Arkansas, Fayetteville, Arkansas, USA*

This supplement documents secondary analyses, illustrative software cases, and reproducibility details that are useful for audit but not necessary to the main argument. It must be read with the evidence classifications in the manuscript.

S1. Evidence inventory and provenance

The repository source registry records each provider-native file, version, access URL, declared license or redistribution constraint, local filename, and SHA-256 checksum. Raw files with uncertain redistribution rights remain under `private-data/` and are excluded from Git. Derived, non-provider tables are written to both `paper/tables/` and a results directory.

The checksum ledger contains 40 provider files in 10 source groups, including all nine historical eGRID downloads. The applied-energy metadata separately records the 11 files that enter its numerical results, their byte counts and SHA-256 hashes, the analysis-script hash, the base seed, and iterations per design. This narrower execution manifest prevents a complete download catalog from being mistaken for the actual computational dependency set.

The manuscript analysis uses the following provider fields:

- Microsoft/WSP: normalized GHG, primary-energy, and blue-water contribution tables; detailed GaBi electricity endpoints; pedigree assessment.

- eGRID: `STNGENAN`, `STCO2AN`, `STCH4AN`, `STN2OAN`, and `STC2ERTA` at the state level and corresponding `US*` fields at the U.S. level.

- Boavizta: product category, manufacturer, product name, report date, lifetime, total GWP, and manufacturing share.

- ÖKOBAUDAT: UUID, product, geography, declared unit, modules A1-A3, GWP, nonrenewable primary energy, and freshwater.

USLCI, GLAD hydrogen, and USGS records are registered and exchange-tested but do not enter a reported LCIA total.

S2. Reproducibility table map

Table S1: Supplementary result map

File	Content	Evidence interpretation
`table1_microsoft_reproduction_audit.csv`	24 released totals and reconstruction error	Arithmetic consistency only
`table2_grid_reductions_vs_air.csv`	Relative reductions from the released model	Reconstructed released result

Continued on next page

Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
`table6_state_rebased_ghg.csv`	2023 state screening outputs	Boundary-mismatched screening transformation
`table8_crossover_thresholds.csv`	Pairwise crossover values	Conditional on a numerical transfer between the two released electricity endpoints
`table9_boavizta_server_records.csv`	48 public product records	Heterogeneous secondary evidence
`table11_material_decarbonization_levers.csv`	Selected matched process comparisons	Per-unit procurement leverage only
`table14_egrid_historical_state_factors.csv`	459 harmonized state-year rates	Direct generation-rate scenarios
`table15_egrid_historical_national_factors.csv`	Harmonized and provider U.S. series	Historical trend and scope audit
`table21_anchor_extrapolation_diagnostic.csv`	Within/below/above anchor counts	Transformation-validity diagnostic

Continued on next page

Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
`table22_boundary_mismatch_stress.csv`	Additive allowance cases	Sensitivity, not an upstream estimate
`table23_joint_assumption_stress.csv`	Seeded stress frequencies	Not probability or confidence
`table24_functional_unit_sensitivity.csv`	Impact-per-service break-even correction	Measurement-resolution target
`table25_priority_index_sensitivity.csv`	Rank stability across index weights	Heuristic diagnostic
`table26_stress_structure_sensitivity.csv`	Assumption-block and dependence designs	Triangular-marginal sensitivity; not empirical probability

S3. Released primary-energy and blue-water results

The 24-cell reconstruction includes GHG, primary energy, and blue water. The main manuscript focuses on GHG because the historical eGRID extension supplies gas-specific air-emission data, not a lifecycle primary-energy or electricity-mediated water inventory.

At the released U.S. grid endpoint, the Microsoft/WSP normalized results show that cold plate, single-phase immersion, and two-phase immersion reduce the three indicators relative to air cooling. These results inherit the released functional unit, boundaries, inventories, allocation, performance, and electricity cases. They were not re-based with eGRID for primary energy or water.

S4. Pairwise crossover screen

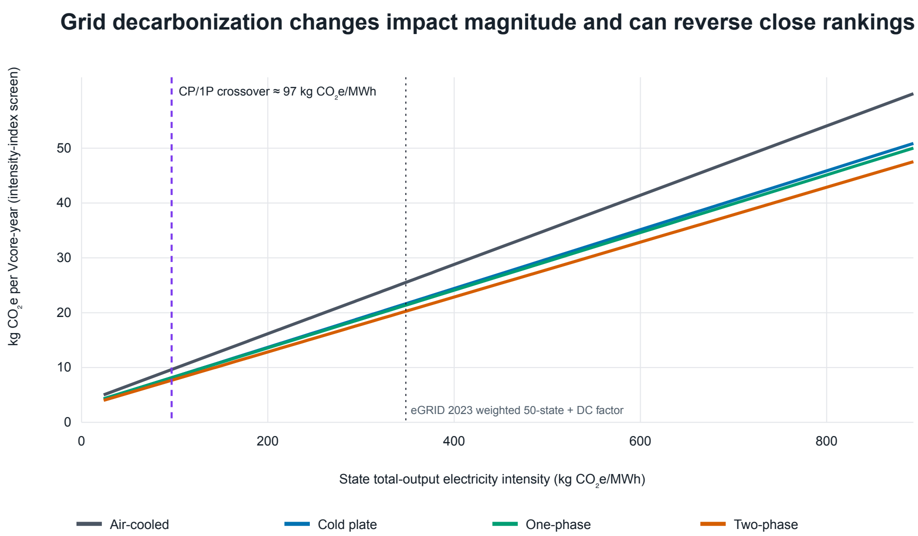

Figure S1: Screening totals transferred across 2023 eGRID state/DC rates. The vertical line is the cold-plate/single-phase crossover. The x-axis is a numerical electricity-intensity index across incompatible lifecycle boundaries, not a background-process substitution.

The two-phase architecture has no crossover with cold plate or single-phase inside the 2023 state-rate range under the fixed released foreground. This absence is conditional on server performance, fluid, equipment quantity, lifetime, and the electricity transfer model.

S5. Deterministic performance discrimination

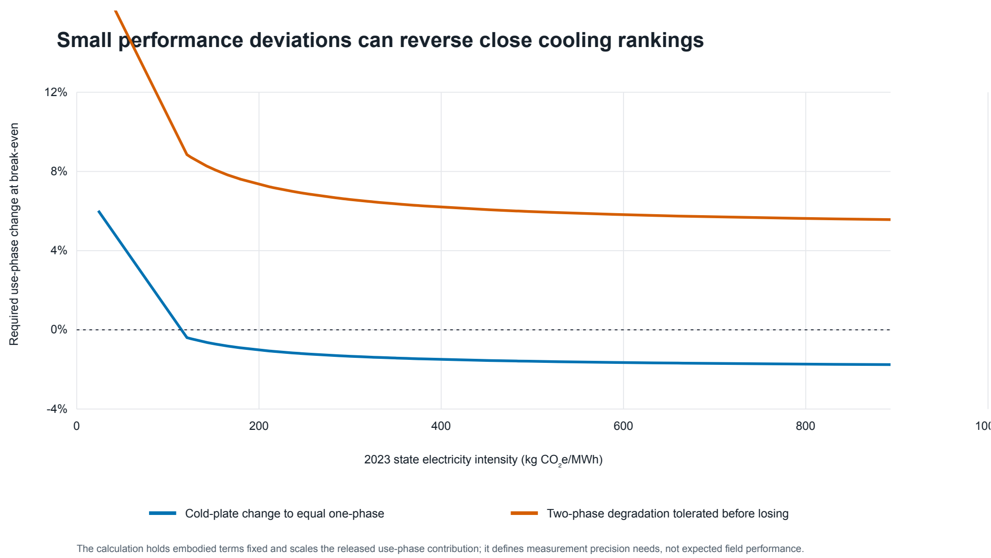

Figure S2: One-at-a-time use-phase change required to alter close rankings across 2023 state rates. These thresholds define experimental discrimination requirements and are not uncertainty intervals.

The cold-plate/single-phase order can be reversed by small use-phase changes. The two-phase margin is larger in the one-at-a-time test but becomes less decisive under joint stress, as reported in the main manuscript.

S6. Boavizta server-product dispersion

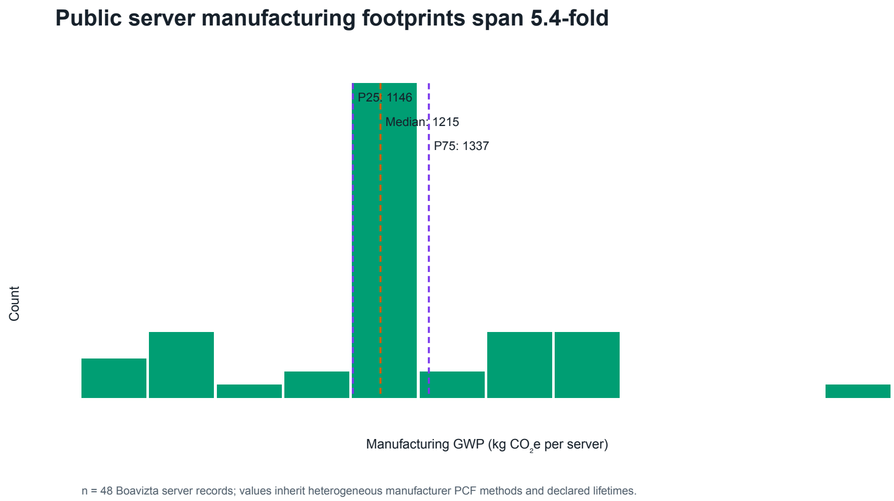

Figure S3: Distribution of manufacturing GHG across 48 Boavizta server records. The records inherit heterogeneous product rules, hardware configurations, lifetimes, and source methods.

The common-scaling stress applies the empirical annualized dispersion to the released compute, storage, and networking contributions simultaneously. It is not a product substitution model. Decision-grade use requires architecture-specific server count, configuration, useful computation, and lifetime under harmonized product-carbon-footprint rules.

S7. ÖKOBAUDAT material-route screen

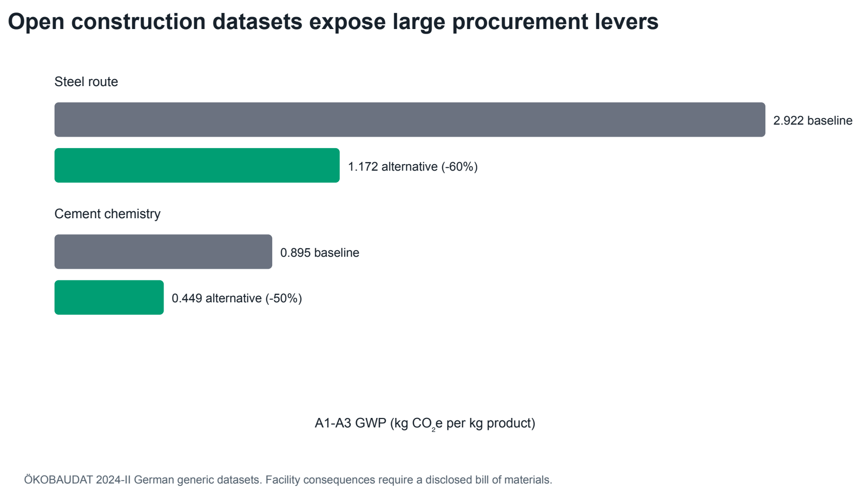

Figure S4: Selected matched ÖKOBAUDAT A1-A3 GHG factors. The comparison identifies per-kilogram procurement leverage and is not propagated to a data-center total.

The selected records are German generic datasets. A facility-level analysis requires architecture-specific quantities, material grade and supplier, fabrication, transport, replacement, and end-of-life alignment.

S8. Hourly climate and performance-map examples

The Fayetteville TMY file contains 8,760 hourly weather records. The repository uses it with a synthetic cooling-performance surface to test:

- schema and unit validation;
- bounded bilinear interpolation;

- uncovered-domain rejection;
- hourly PUE and operational-impact integration; and
- CLI, API, GUI, and report consistency.

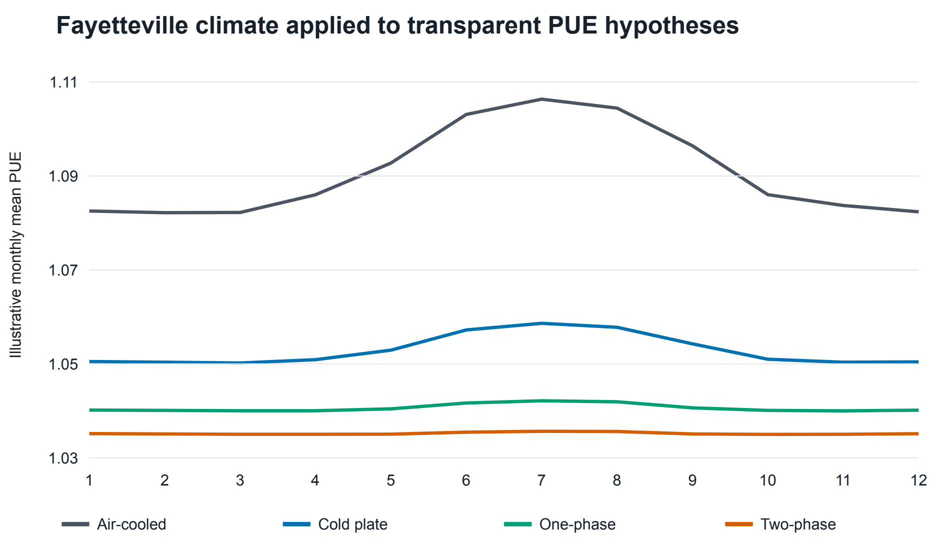

Figure S5: Monthly PUE from the Fayetteville weather and declared illustrative performance surface. This is a software example, not empirical evidence for a cooling technology.

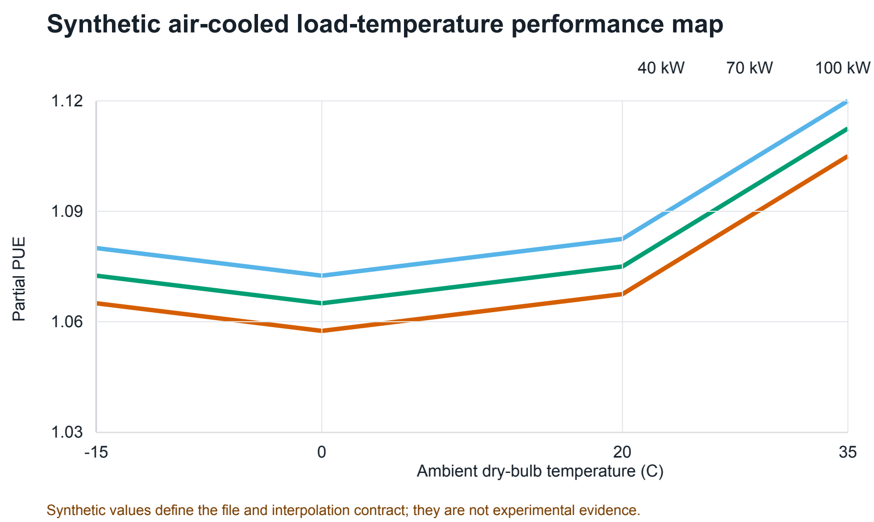

Figure S6: Synthetic load-temperature cooling surface used to verify bounded interpolation. No comparative assertion is permitted from this fixture.

Submission-grade energy results require architecture-specific measured power for IT-integral fans, pumps, coolant distribution, heat rejection, controls, and onsite water over a common load-weather domain.

S9. Data-quality priority and its limitations

The base priority index multiplies mean released GHG contribution by normalized pedigree weakness. Alternative exponents show that use phase is usually, but not invariably, first. Storage remains consistently in the top tier. This robustness test prevents the heuristic index from being described as a formal value-of-information result.

Formal value of information would additionally require:

- empirically defensible parameter distributions and correlations;

- the probability and consequence of selecting the wrong architecture;
- cost, duration, and achievable precision of each measurement or inventory;
- a decision threshold and stakeholder utility; and
- posterior updating after data acquisition.

The main-text block decomposition complements this heuristic. Under the declared U.S. wide envelope, embodied-only variation reduced the two-phase first-rank frequency to 63.36%, compared with 79.52% for use only, 73.90% for service only, and 100% for grid only. These frequencies rank leverage only within the declared stress widths; they do not estimate empirical variance.

`table27_stress_convergence.csv` records nested 5,000-, 20,000-, and 80,000-draw runs. For the U.S. narrow, screening, and wide envelopes, the 20,000-draw estimates differed from the 80,000-draw estimates by 0.058, 0.056, and 0.110 percentage points, respectively. The largest difference across both locations was 0.709 percentage points. The table reports numerical Monte Carlo standard errors but does not interpret them as empirical uncertainty.

S10. Reliability and replacement

The package supports discrete replacement, linearized replacement, Weibull renewal expectation, and optional Arrhenius acceleration. These functions are tested independently of the Microsoft/WSP secondary analysis.

They do not provide architecture-specific reliability without failure and maintenance data.

Required records include component definition, population, duty cycle, temperature history, failure mode, censoring, repair/replacement action, downtime, spare inventory, fluid loss, and calibration/inspection history. Dependent failures, redundancy, and repair queues require an extended model.

S11. Interoperability boundary

Successful openLCA JSON-LD parsing or Brightway mapping establishes syntactic exchange. Numerical compatibility additionally requires matching:

- reference product and unit;
- provider and database version;
- geography and reference year;
- attributional or consequential system model;
- allocation and cut-off;
- elementary-flow nomenclature;
- LCIA method and version; and
- foreground product-system linkage.

The seven GLAD hydrogen archives and USLCI database are therefore preserved as future inputs, not silently substituted into the current cooling results.

S12. Reproduction commands

From the repository root:

```
python -m pip install -e .
python scripts/build_source_manifest.py --check
python scripts/run_integrated_evidence_analysis.py
python scripts/run_applied_energy_analysis.py
python -m unittest discover -s tests -v
python paper/build_manuscript.py
python paper/build_overleaf.py
```

The scripts use fixed output paths, newline-stable CSV writing, and stable scenario-specific seeds. A clone without `private-data/` runs the public package tests and explicitly skips eight paper-reconstruction tests. Rebuilding the paper requires the locally held provider files whose hashes appear in the manifest. The manifest, derived tables, analysis metadata, and exact Git commit should be archived together for submission.